

How K-Means Clustering Works

K-means is an iterative algorithm that automatically groups unlabeled data into clusters. It works by repeatedly performing two specific steps until it finds the best grouping.

The Algorithm

1. Initialization:

- The algorithm starts by taking a **random guess** at where the centers of the clusters might be.
- It picks K random points (where K is the number of clusters you want to find). These points are called **Cluster Centroids**.

2. The Iterative Loop:

K-means repeatedly cycles through two main steps:

- **Step 1: Assign points to cluster centroids**
 - The algorithm goes through every data point ($x^{(1)}$ to $x^{(m)}$).
 - For each point, it calculates the distance to the centroids.
 - It assigns (or "colors") the point to the cluster centroid it is **closest to**.
- **Step 2: Move cluster centroids**
 - The algorithm looks at all the points assigned to a specific cluster (e.g., all the red points).
 - It calculates the **average (mean)** location of those points.
 - It moves the centroid to that new average location.

Convergence

- After moving the centroids (Step 2), the algorithm goes back to Step 1 and re-assigns points based on the *new* centroid locations. Some points might change clusters (change color) because the centroids have moved.
- This cycle (Assign → Move → Assign → Move) repeats.
- **Convergence** occurs when there are no more changes:
 - Points stop switching clusters (colors remain the same).
 - Centroids stop moving (their location is stable).

At this point, the algorithm has successfully grouped the data into distinct clusters.